

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
11.	(a)	(i)		any three of following for (1) each - needs yeast / enzyme added to the mixture - should be kept at body temperature / 25-40°C - must have some means of allowing CO₂ produced to escape / keeping oxygen out - distillation needed to separate the ethanol	1 1	1		3		3
		(ii)		180 g of glucose produces 92 g of ethanol (1) atom economy = $92/180 \times 100 = 51.1\%$ (1)		2		2	1	
		(iii)		hydration of ethene needs high temperature / lots of energy (1) this uses fossil fuels (1) OR hydration of ethene uses non-renewable petroleum (1) fermentation uses renewable sugars / plant materials / is carbon neutral (1)	1		1	2		